

CHILD GROWTH ABNORMALITY DETECTION FOR AUTISTIC PATIENT USING VISION TRANSFORMER

PROJECT REPORT

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ABSTRACT

Autism Spectrum Disorder is a complex neurodevelopmental condition that affects the way an individual communicates, behaves, and interacts with others. Early detection plays a very important role in improving the developmental outcomes of affected individuals, but traditional diagnostic methods rely heavily on clinical observation and behavioral assessments, making the process time-consuming, subjective, and resource-intensive. This project proposes an automated deep learning based system for the detection of Autism Spectrum Disorder using facial image analysis. The proposed system uses a Vision Transformer architecture that processes facial images by dividing them into fixed size patches and learning the global relationships between all patches simultaneously using a multi-head self-attention mechanism. Unlike traditional Convolutional Neural Network models that focus only on local regions of the image, the Vision Transformer captures global facial relationships, allowing it to identify subtle visual patterns associated with Autism Spectrum Disorder more effectively. To improve interpretability, an RGB Attention Rollout Visualization technique is integrated into the system, which highlights the important facial regions using a color coded heatmap where red indicates high attention, green indicates medium attention, and blue indicates low attention. The proposed system offers a non-invasive, cost-effective, and interpretable solution for early Autism Spectrum Disorder detection that can be used as a supportive tool for medical professionals in real-world clinical settings.

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LIST OF ABBREVIATION

ML	Machine Learning
DL	Deep Learning
ATM	Attention Rollout Mechanism
ASD	Autism Spectrum Disorder
VIT	Vision transformer
FF	Feed Forward
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
F1	F1 Score
ANN	Artificial Neural Network
RGB	Red Blue Green
CLS	Classification Token
MHA	Multi- Head Attention
MLP	Multi-Layer Perceptron

CHAPTER 1

INTRODUCTION

1.1 MACHINE LEARNING

Machine Learning is an important branch of Artificial Intelligence that allows computers to learn from data and make decisions without being explicitly programmed for every task. In traditional programming, instructions are given step by step to solve a problem, but in machine learning, the system learns patterns from the data and uses those patterns to predict outcomes. This makes machine learning more flexible and efficient when dealing with real-world problems where it is difficult to define exact rules. The main idea behind machine learning is that the system improves its performance as it is exposed to more data over time. It works by training a model using a set of input data and corresponding outputs, which helps the system understand the relationship between them. Once the training is completed, the model can be used to make predictions on new and unseen data.

Machine learning is widely used in many applications such as image recognition, speech recognition, recommendation systems, and medical diagnosis. For example, in image recognition, the model learns to identify objects by analyzing thousands of images. Similarly, in healthcare, machine learning models are used to detect diseases by analyzing medical data and images. This makes the process faster and more accurate compared to manual methods. There are different types of machine learning such as supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, the model is trained using labeled data, where the correct output is already known. In unsupervised learning, the model finds patterns in data without any labels. Reinforcement learning is a method where the system learns by interacting with the environment and receiving feedback in the form of rewards or penalties.

One of the major advantages of machine learning is its ability to handle large amounts of data and find hidden patterns that are not easily visible to humans. It reduces human effort and increases efficiency in many applications. However, machine learning also has some challenges such as the need for large datasets, proper training, and the risk of overfitting if the model is not designed correctly. Despite these challenges, machine learning continues to play a major role in modern technology. In this project, machine learning forms the base for building an intelligent system that can detect Autism Spectrum Disorder using facial images, making the process faster and more reliable.

Machine learning systems are known for operating in a black box, meaning you have no visibility into how the machine learns and makes decisions. Thus, if you identify an instance of bias, there is no way to identify what caused it. Your only recourse is to retrain the algorithm with additional data, but that is no guarantee to resolve the issue. ML provides the foundation for AI systems that automate processes and solve data-based business problems autonomously. It enables companies to replace or augment certain human capabilities.

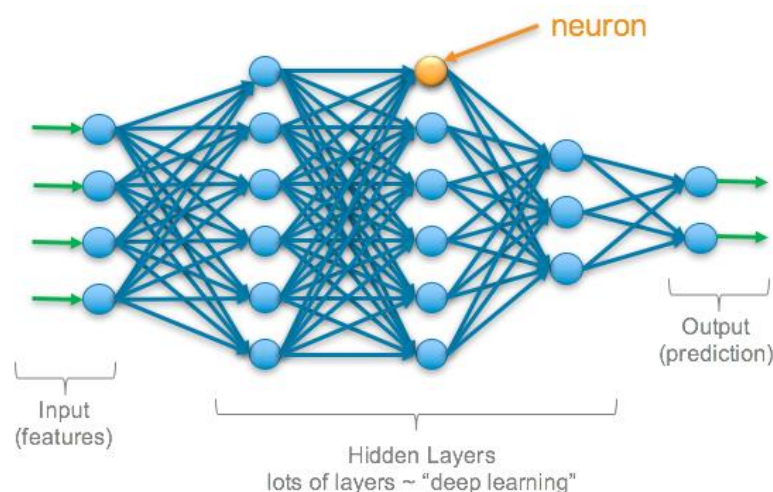


Figure 1.1 Machine Learning Architecture

Common machine learning applications you may find in the real world

include chatbots, self-driving cars and speech recognition , Traffic prediction,Product recommendations,Self-driving cars, Email Spam and Malware Filtering, Virtual Personal Assistant,Online Fraud Detection.

1.2 DEEP LEARNING

Deep Learning is a subfield of machine learning which involves algorithms that are inspired from working functions of the biological brain. Deep Learning (DL) involves replicating human neurons by establishing a system of nodes which act as a single unit of neuron. A standard neural network is called artificial neural networks which consists of nodes and weights upon different layers. There are three types of foundational node layers which are input nodes where the input data is passed, Output nodes where the output prediction is seen and the hidden layer in between the two which find the patterns in the input data.

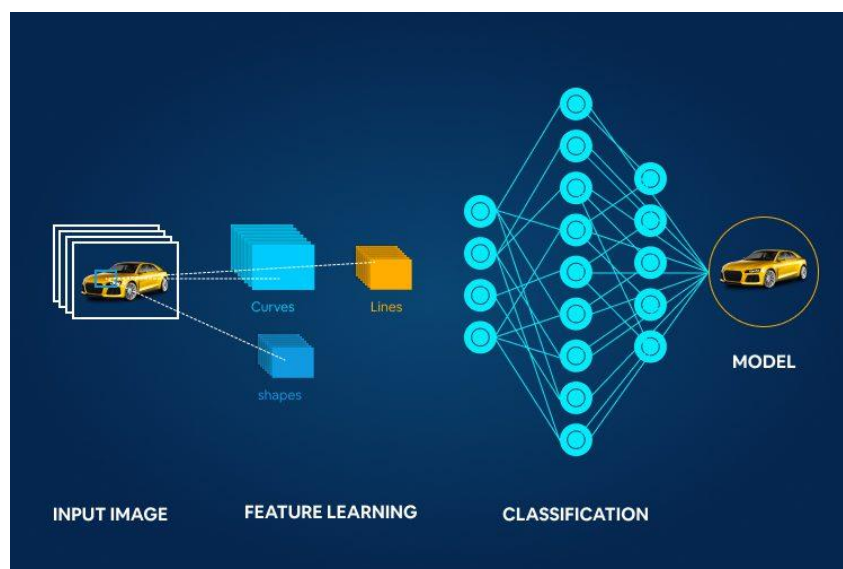


Figure 1.2 Deep Learning Architecture

Each node has assigned a weight bias initially upon which the neural network is trained with the existing dataset. Using the necessary feedback. The

weights between the nodes are adjusted. This process is called backpropagation. The most efficient weight combination is found using a gradient descent technique which involves a multi dimensional graph comparing between weights and the output cost function. The output cost function is determined by a

function of difference between desired output with respect to the existing output. Using multivariable calculus, The efficient local minima is found where the cost function is significantly low according to which the weights are adjusted . This process of back propagation is iterative and happens like a chain process .Deep-learning architectures have been applied to fields including computer vision, speech recognition, natural language processing, machine translation, bioinformatics, drug design, medical image analysis and material inspection etc.

Different types of neural networks are used for different data and applications. The different architectures of neural networks are specifically designed to work on those particular types of data or domain. Let's start from the most basic ones and go towards more complex ones.

In deep learning, the model automatically learns features from the data without the need for manual feature extraction. This is one of the major advantages compared to traditional machine learning methods. For example, in image processing, the model can learn edges, shapes, textures, and patterns directly from the images. This makes deep learning very powerful for tasks like image classification, object detection, and facial recognition. It has been widely used in many applications such as self-driving cars, voice assistants, medical image analysis, and natural language processing.

Deep learning models are usually built using different types of neural networks. Some of the commonly used networks include Convolutional Neural Networks (CNN), Recurrent

Neural Networks (RNN), and Transformer models. CNN is mainly used for image-related tasks because it can capture spatial features effectively. RNN is used for sequential data such as text and speech. Recently, transformer-based models have gained popularity because they can capture relationships in data more efficiently.

Even though deep learning provides high accuracy, it also requires a large amount of data and computational power. Training deep learning models can take a long time, especially when the dataset is large. It also requires proper tuning of parameters to achieve better performance. Despite these challenges, deep learning has become one of the most important technologies in modern artificial intelligence. In this project, deep learning plays a key role in building a system that can analyze facial images and detect Autism Spectrum Disorder with improved accuracy and efficiency.

Deep learning models improve gradually as they are trained with more data, allowing them to achieve better accuracy over time. They are capable of solving complex problems where writing exact rules is difficult. One of the main advantages is that they reduce human effort by automatically learning important features from the data. These models can work with different types of data such as images, text, audio, and video. However, training deep learning models requires powerful hardware like GPUs and can take a significant amount of time. Once the training is complete, the model can make predictions very quickly. Sometimes, deep learning models behave like a “black box,” meaning it is difficult to understand how they arrive at a decision. They may also suffer from overfitting if they learn too much from the training data and fail to perform well on new data. Therefore, proper testing and good quality data are very important to ensure reliable results.

1.3 CONVOLUTIONAL NEURAL NETWORKS

Convolutional neural network is a class of artificial neural network (ANN) which can take in an input image, assign learnable weights and biases to various aspects/objects in the image and be able to differentiate one from the other.

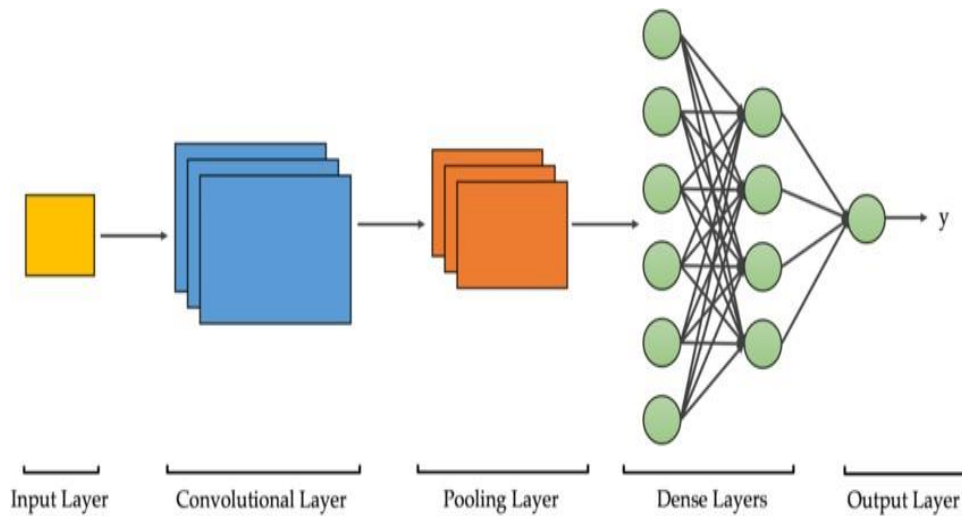


Figure 1.3 Basic Architecture of Convolutional Neural Network

Convolutional Neural Network (CNN) is a class of Artificial Neural Network (ANN) which can take in an input image, assign learnable weights and biases to various aspects/objects in the image and be able to differentiate one from the other. CNNs are also known as Shift Invariant or Space Invariant Artificial Neural Networks (SIANN), based on the shared-weight architecture of the convolution kernels or filters that slide along input features and provide translation-equivariant responses known as feature maps.

Convolution is a mathematical operation that allows the merging of two sets of information. In the case of CNN, convolution is applied to the input data to filter the information and produce a feature map. This filter is also called a kernel, or feature detector, and its dimensions can be, for example, 3×3 . To

perform convolution, the kernel goes over the input image, doing matrix multiplication element after element. The result for each receptive field (the area where convolution takes place) is written down in the feature map.

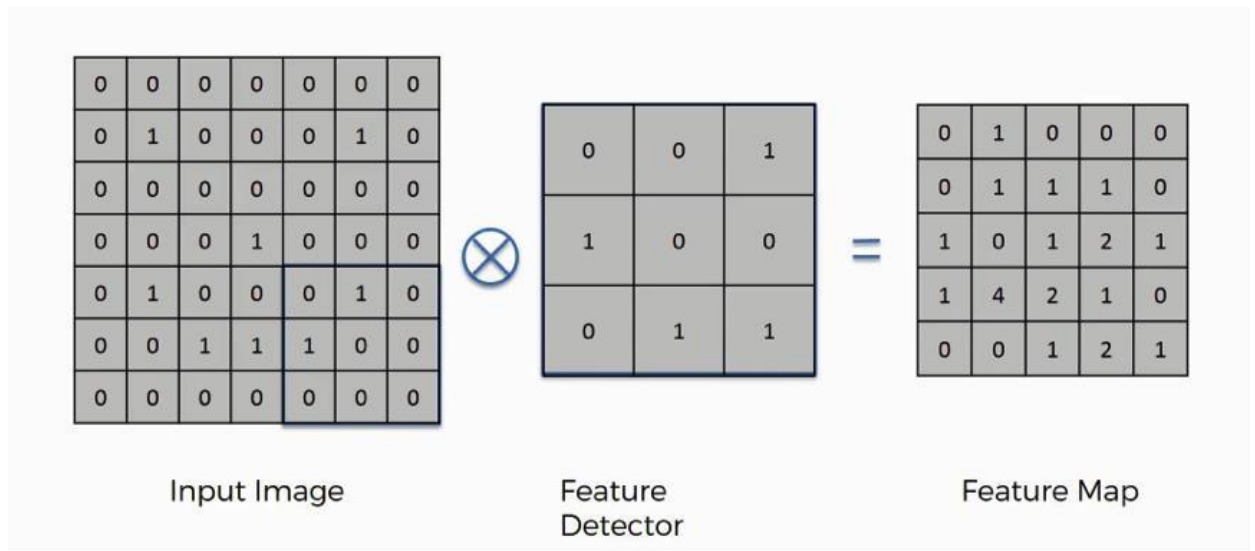


Figure 1.3 Convolution Function

There are several uses that is gained from deriving a feature map. These are the most important of them: Reducing the size of the input image, and you should know that the larger your strides, the smaller your feature map. In this example, we used one-pixel strides which gave us a fairly large feature map. When dealing with proper images, you will find it necessary to widen your strides. Here we were dealing with a 7×7 input image after all, but real images tend to be substantially larger and more complex.

Convolutional Neural Network were inspired by the working of neurons in the brain where each neuron acts as a node which transmits an electrical pulse via which the brain functions. A Convolutional Neural Network (CNN) uses a system much like a multilayer perceptron that has been designed for reduced processing requirements. The layers of a Convolutional Neural Network (CNN)

consist of an input layer, an output layer and a hidden layer that includes multiple convolutional layers, pooling layers, fully connected layers and normalisation layers. Convolutional Neural Network is predominantly used in image recognition since images can be converted to a sequence of matrices and passed as an input. They have applications in image and video recognition, image classification, medical image analysis, natural language processing and brain–computer interface. The main advantage of CNN compared to its predecessors is that it automatically detects the important features without any human supervision. CNN has been used in a variety of real-world applications like cancer detection and biometric authentication. CNN can also be applied to visual question answering or image captioning where CNN networks take an input image and generate natural language answers about that image.

1.4 VISION TRANSFORMER

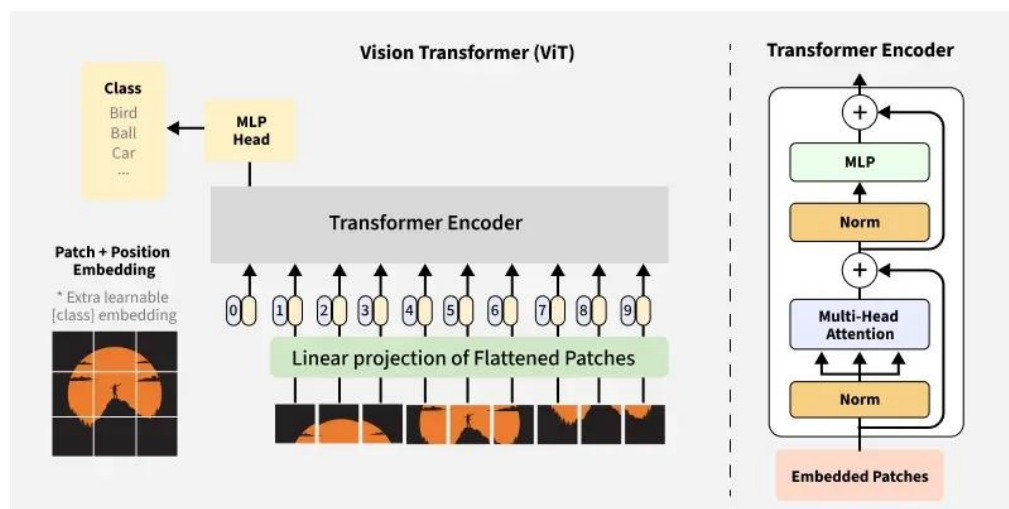


Figure 1.4 Vision Transformer Architecture

Vision Transformer, commonly referred to as ViT, is a deep learning model that applies the concept of transformers to image data. Transformers were originally designed for natural language processing tasks, where they proved to be highly effective in understanding the relationship between words in a sentence. Researchers later discovered

that the same attention-based mechanism used in language models could also be applied to images. This led to the development of the Vision Transformer, which processes images in a completely different way compared to traditional convolutional neural networks.

Instead of scanning the image using filters, the Vision Transformer divides the image into small fixed-size patches and processes all these patches together using a mechanism called self-attention. This approach allows the model to understand the relationship between different parts of the image at the same time, which makes it more powerful for complex image recognition tasks. The working of a Vision Transformer begins with dividing the input image into equal-sized patches.

For example, an image of size 224×224 pixels may be divided into patches of size 16×16 , resulting in a total of 196 patches. Each patch is then flattened into a one-dimensional vector and converted into a fixed-size embedding using a linear projection layer. These patch embeddings carry the visual information from each region of the image. Along with the patch embeddings, positional encoding is added to each patch so that the model knows the position of each patch within the original image. Without positional encoding, the model would not be able to understand the spatial arrangement of the patches, which is important for recognizing meaningful patterns in the image.

Once the patch embeddings are prepared, a special token called the classification token or CLS token is added at the beginning of the sequence. This token does not correspond to any specific patch but acts as a summary representation of the entire image. The entire sequence of patch embeddings along with the CLS token is then passed through multiple transformer encoder layers. Each encoder layer contains a multi-head self-attention mechanism and a feed-forward neural network. The self-attention mechanism computes the relationship between every patch and every other patch in the image, allowing the model to focus on the most relevant regions when making predictions. This global understanding of the image is one of the key advantages of the Vision Transformer over convolutional neural networks.

After passing through all the transformer encoder layers, the output corresponding to the CLS token is extracted and passed through a classification head, which is usually a fully connected layer. This final layer maps the learned representation to the output classes, which in this project are Autism Spectrum Disorder and Non-ASD. The Vision Transformer has shown excellent performance on various image classification benchmarks and has proven to be more effective than traditional models in tasks that require understanding of global features. In this project, the Vision Transformer is used to analyze facial images and capture important facial patterns that help in detecting Autism Spectrum Disorder with high accuracy and reliability.

1.5 AUTISM SPECTRUM DISORDER

Autism Spectrum Disorder, commonly known as ASD, is a complex neurodevelopmental condition that affects the way a person communicates, behaves, and interacts with others. It is called a "spectrum" disorder because the symptoms and their severity vary widely from one individual to another. Some individuals with autism may have difficulty in social communication and prefer to spend time alone, while others may have strong abilities in certain areas such as mathematics, music, or art. The condition usually appears in early childhood, often before the age of three, and continues throughout the person's lifetime.

Since autism affects people in different ways, it is not always easy to identify, and many cases go undetected until later stages of development. Early identification and intervention are considered extremely important because timely support can significantly improve the quality of life and developmental outcomes for individuals with autism.

The exact cause of Autism Spectrum Disorder is still not fully understood by the medical community. Research suggests that it is caused by a combination of genetic and environmental factors. Studies have shown that certain genes play a role in the development of autism, and it tends to run in families. Environmental factors such as exposure to certain chemicals during pregnancy, complications during birth, and advanced parental age have also been associated with a higher risk of autism. However, no single factor has been identified as the definitive cause. It is believed that the condition results from differences in the way the brain develops and functions, particularly in areas responsible for social interaction and communication. Brain imaging studies have revealed that individuals with autism show different patterns of brain activity compared to those without the condition.

The symptoms of Autism Spectrum Disorder are generally observed in three main areas, which are social interaction, communication, and repetitive behavior. In terms of social interaction, individuals with autism may find it difficult to make eye contact, understand facial expressions, or respond appropriately to social situations. They may not be able to recognize the emotions of others or express their own emotions in a typical way. In communication, some individuals with autism may have delayed speech development, while others may have difficulty understanding the meaning behind words or gestures. Repetitive behaviors are also commonly seen, such as repeating specific actions, following strict routines, or showing intense interest in a narrow range of subjects. These symptoms can vary greatly in type and intensity from one person to another, making diagnosis a challenging process.

Diagnosing Autism Spectrum Disorder traditionally involves detailed observation by trained specialists, behavioral assessments, and interviews with parents or caregivers. There are no specific medical tests such as blood tests or brain scans that can directly confirm the diagnosis. The process is time-consuming and requires highly skilled professionals, making

it difficult to reach all individuals who may need assessment, especially in areas with limited medical resources. This is one of the major challenges in autism diagnosis across the world. In recent years, researchers have been exploring the use of artificial intelligence and deep learning to support and speed up the diagnostic process. Facial image analysis using advanced deep learning models has shown promising results, as certain facial features and expressions may carry useful information related to autism. This project builds on that idea by using a Vision Transformer model to analyze facial images and assist in early detection of Autism Spectrum Disorder in a faster, more objective, and scalable manner.

1.6 ATTENTION MECHNAISM

The attention mechanism is a fundamental concept in deep learning that allows a model to focus on the most relevant parts of the input data when making a prediction or decision. It was originally introduced in the field of natural language processing to help models handle long sequences of text more effectively. Before the attention mechanism was developed, models processed input data in a sequential manner, which made it difficult to retain important information over long distances. The attention mechanism solved this problem by allowing the model to look at all parts of the input at the same time and assign different levels of importance to each part based on its relevance to the current task. This ability to selectively focus on important information has made the attention mechanism one of the most powerful tools in modern deep learning, and it has been successfully applied to a wide range of tasks including image classification, object detection, machine translation, and medical image analysis.

The basic idea behind the attention mechanism can be understood through a simple analogy. When a person reads a sentence to answer a question, they do not pay equal attention to every word. Instead, they focus more on the words that are most relevant to the question. Similarly, when a person looks at an image, they naturally focus on the most

important regions rather than analyzing every pixel with equal effort. The attention mechanism in deep learning works in a similar way. It computes a score for each part of the input, which represents how much attention should be given to that part. These scores are then converted into weights using a softmax function, and the weighted combination of all input parts is computed to produce the final output.

This process allows the model to dynamically decide which parts of the input are most useful for each specific prediction.

In the context of Vision Transformers, the attention mechanism used is called multi-head self-attention. In self-attention, the model computes the relationship between every patch of the image and every other patch. For each patch, three vectors are computed which are called the Query, Key, and Value vectors. The Query vector represents what the current patch is looking for, the Key vector represents what each patch has to offer, and the Value vector contains the actual information of each patch. The attention score between two patches is computed by taking the dot product of the Query vector of one patch with the Key vector of another patch. These scores are then scaled and passed through a softmax function to obtain attention weights. The final output for each patch is computed as a weighted sum of all Value vectors, where the weights are the attention scores. This process is repeated multiple times in parallel using different sets of Query, Key, and Value matrices, which is why it is called multi-head attention. Each head learns to focus on different aspects of the image, and the outputs of all heads are combined to produce a rich representation of the input.

The attention mechanism plays a very important role in this project because it allows the Vision Transformer model to identify and focus on the most relevant facial regions when detecting Autism Spectrum Disorder. Instead of treating all parts of the face equally, the model learns to pay more attention to regions such as the eyes, nose, and mouth, which may carry important information related to autism. This selective focus improves the accuracy of the model and also makes it more interpretable. By visualizing the attention scores, it is possible to understand which parts of the face the model is focusing on when making a

prediction. This transparency is especially important in medical applications where doctors and clinicians need to trust and understand the decisions made by the model before using it in real-world diagnosis.

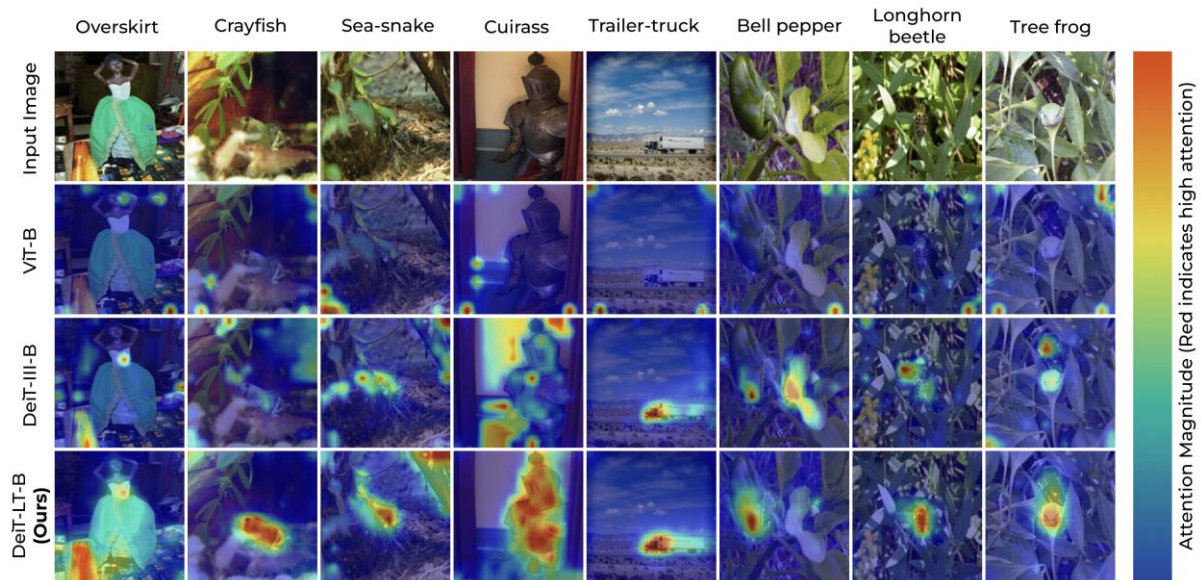


Figure 1.5 Attention Rollout Mechanism

1.7 COMPLEX ENGINEERING PROBLEM ADDRESSED

The complex engineering problem addressed in this project is the automated detection of Autism Spectrum Disorder from facial images using deep learning, specifically the Vision Transformer architecture. Autism Spectrum Disorder is a neurodevelopmental condition that affects communication, behavior, and social interaction, and its diagnosis has traditionally relied on clinical observation and behavioral assessments conducted by trained specialists. This conventional approach is highly time-consuming, subjective, and resource-intensive, making it inaccessible to large populations, particularly in regions with limited medical infrastructure. The challenge lies in building a system that can analyze subtle and complex visual patterns in facial images that are associated with ASD, which are not easily identifiable through manual observation or conventional image processing techniques.

The engineering complexity of this problem arises from multiple conflicting demands that must be satisfied simultaneously. The system must achieve high classification accuracy while working with a very small dataset of only 280 labeled facial images, which creates a significant risk of overfitting. At the same time, the model must generalize well across variations in age, gender, ethnicity, lighting, and facial expression. Traditional Convolutional Neural Networks were found to be insufficient because they only capture local spatial features, whereas ASD-related facial patterns require understanding of global relationships across the entire face. This necessitated the adoption of the Vision Transformer, which processes all image patches simultaneously through a multi-head self-attention mechanism, but introduced further complexity in terms of computational requirements and the need for transfer learning from large-scale pretraining.

Beyond accuracy, the problem also demanded that the system be interpretable and trustworthy enough for use in real-world clinical settings, since medical professionals cannot rely on a black-box model for diagnostic support. This added an additional engineering challenge of integrating an RGB Attention Rollout Visualization technique that extracts and combines attention weights from all transformer encoder layers to generate a color-coded heatmap, highlighting the facial regions most relevant to the model's prediction. The system therefore had to be designed as an end-to-end solution that is not only accurate but also explainable, scalable, and deployable with limited hardware resources, making it a genuinely complex engineering problem that required careful balancing of multiple technical and clinical constraints.

1.8 SUSTAINABLE DEVELOPMENT GOAL (SDG) MAPPING



Figure 1.8 Sustainable Development Goals (SDG) Mapping

SDG 3 – Good Health and Well-Being: This project directly contributes to SDG 3 by addressing one of the most significant challenges in neurodevelopmental healthcare, which is the early and accurate detection of Autism Spectrum Disorder. By developing an automated deep learning system that analyzes facial images to classify individuals as autistic or non-autistic with approximately 91.7% accuracy, the project promotes timely diagnosis and intervention for children affected by ASD. Early detection is clinically proven to significantly improve developmental outcomes, quality of life, and the overall well-being of affected individuals. The system serves as a cost-effective and non-invasive supportive tool for medical professionals, reducing dependence on lengthy and subjective clinical assessments and making healthcare more proactive and accessible for all age groups.

SDG 10 – Reduced Inequalities: One of the core limitations of traditional ASD diagnosis is that it requires highly trained specialists and expensive clinical procedures, making it largely inaccessible to communities in low-resource or rural settings. This project addresses that inequality by proposing a scalable, automated screening system that can function with standard hardware such as an Intel i5 processor and 8GB RAM, without the need for costly medical infrastructure. Furthermore, the dataset used in training includes facial images across diverse age groups, genders, and ethnicities, ensuring the model is inclusive and does not favor any particular demographic. By democratizing access to early

ASD screening, the project works toward reducing health-related inequalities within and across communities.

SDG 17 – Partnerships for the Goals: The development of this system reflects the spirit of SDG 17 by integrating knowledge and tools from multiple domains including artificial intelligence, medical science, and data science to achieve a shared healthcare goal. The project leverages publicly available datasets from Kaggle, pretrained models from large-scale research initiatives like ImageNet, and open-source frameworks such as PyTorch and TorchVision, all of which represent collaborative contributions from the global research and technology community. The interpretability feature through RGB Attention Rollout Visualization also fosters trust and collaboration between AI developers and clinical professionals, creating a bridge between technology and medicine that is essential for sustainable and impactful implementation in real-world healthcare systems.

CHAPTER 2

LITERATURE REVIEW

Bhatt et al. (2020) conducted a study on the use of deep learning models for detecting Autism Spectrum Disorder using facial images. The main idea behind their work was that facial images of individuals with autism may carry certain visual patterns that can be identified using deep learning techniques. In the past, autism diagnosis was done entirely through clinical observation and behavioral assessments, which required a lot of time and the involvement of trained specialists. The authors believed that by using deep learning, the process could be made faster and more objective. To achieve this, they collected a dataset of facial images from children who had been diagnosed with autism and from children who were developing typically. The images were carefully labeled and preprocessed before being used for training the model. A Convolutional Neural Network was chosen as the main model for this study because it is well known for its ability to extract meaningful features from image data automatically without the need for manual feature engineering. The CNN model was designed with multiple convolutional layers that extracted features from the facial images at different levels of detail. The earlier layers of the network focused on simple features such as edges and textures, while the deeper layers captured more complex patterns such as the shape of facial regions and the relationship between different parts of the face.

Thabtah et al. (2019) proposed a machine learning based approach for the early detection of autism using behavioural and clinical data collected from individuals of different age groups. The motivation behind this study was that existing autism diagnosis methods were too slow and required the involvement of highly trained medical professionals, making it difficult to reach a large number of people who needed screening. The authors believed that by using machine learning algorithms on easily collectible behavioural data, it would be possible to develop a faster and more accessible screening tool that could be used even in

areas with limited medical facilities. The data used in this study was collected through a standardised autism screening questionnaire that asked questions about the social behaviour, communication patterns, and daily activities of the individuals. The responses were recorded and converted into numerical features that could be used as input to machine learning models. The dataset included individuals from three different age groups, which were toddlers, children, and adults, making the study more comprehensive and applicable to a wider population. Several machine learning algorithms were tested in this study to find the most effective approach for autism classification. The algorithms included Decision Trees, Random Forest, Support Vector Machine, and Naive Bayes classifier. Each algorithm was trained on the collected dataset and evaluated using standard performance metrics such as accuracy, sensitivity, and specificity.

Dosovitskiy et al. (2020) introduced the Vision Transformer model in their research work and demonstrated for the first time that a pure transformer architecture could be successfully applied to image classification tasks without using any convolutional layers. Before this work, convolutional neural networks were considered the standard approach for image recognition, and most researchers believed that the local feature extraction ability of CNNs was essential for handling image data. The authors challenged this assumption by showing that transformers, which were originally developed for natural language processing tasks, could also achieve excellent results on image classification benchmarks when given enough training data. The key motivation behind their work was to explore whether the self-attention mechanism of transformers, which had proven so effective in capturing relationships between words in a sentence, could also be used to capture relationships between different regions of an image. This idea turned out to be very powerful, and the Vision Transformer quickly became one of the most influential models in the field of computer vision.

Sherkatghanad et al. (2020) conducted a research study on the use of deep learning models for detecting Autism Spectrum Disorder using brain imaging data obtained from functional

Magnetic Resonance Imaging scans. The main motivation behind this study was that traditional methods of autism diagnosis were heavily dependent on behavioral observations and clinical interviews, which were subjective in nature and required a significant amount of time and expertise. Functional Magnetic Resonance Imaging, commonly known as fMRI, is a technique that measures brain activity by detecting changes in blood flow in different regions of the brain. When a particular region of the brain is more active, it consumes more oxygen, and the fMRI scan captures this change in blood flow as a signal. By analyzing these signals, it is possible to understand how different regions of the brain communicate with each other, which is known as functional brain connectivity. The dataset used in this study was obtained from the Autism Brain Imaging Data Exchange repository, which is one of the largest publicly available collections of brain imaging data for autism research.

Li et al. conducted a study on child developmental disorder detection using Vision Transformer models applied to facial image datasets. The motivation behind their work was that subtle facial features and expressions could provide early indicators of developmental abnormalities, which are often difficult to detect through manual observation. The authors used a dataset consisting of children's facial images collected from clinical and non-clinical sources. The images were preprocessed using normalization and resizing techniques to ensure consistency. Unlike traditional Convolutional Neural Networks, the Vision Transformer divided each image into smaller patches and processed them as sequences, allowing the model to capture global relationships across the entire face. The self-attention mechanism helped the model focus on important facial regions such as the eyes, mouth, and forehead. The results showed that the Vision Transformer achieved higher accuracy compared to CNN-based models, especially when trained on large datasets, making it a promising approach for early detection of developmental abnormalities.

Khan et al. proposed a hybrid deep learning framework combining Vision Transformer and Convolutional Neural Networks for detecting abnormalities in child growth patterns using medical imaging data. The main idea of this research was to take advantage of both local feature extraction from CNN and global context understanding from Vision

Transformer. The dataset used in this study included medical images such as facial scans and body posture images of children from different age groups. The CNN component extracted low-level features like edges and textures, while the Vision Transformer analyzed relationships between different regions of the image. This combination improved the overall performance of the model. The authors evaluated the model using metrics such as accuracy, precision, and recall, and found that the hybrid model performed better than individual models. This study highlighted the importance of combining different architectures for improving abnormality detection systems.

Park et al. introduced a Vision Transformer-based approach for analyzing skeletal growth abnormalities in children using X-ray images. The motivation behind this work was to develop an automated system that could assist doctors in identifying abnormal bone development at an early stage. The dataset consisted of pediatric X-ray images labeled by medical experts. The images were divided into patches and processed using the transformer architecture, which allowed the model to learn relationships between different bone regions. The attention mechanism helped in identifying important regions such as joints and growth plates. The study showed that the Vision Transformer was able to detect abnormalities with high accuracy and reduced the need for manual analysis. This approach proved useful in supporting early diagnosis and treatment planning.

Chen et al. conducted a study on detecting developmental delays in children using multimodal data, including facial images and behavioral information, with the help of transformer-based models. The authors aimed to improve the accuracy of detection by combining visual and non-visual data. The facial images were processed using a Vision Transformer, while the behavioral data was processed using a separate neural network. The features from both models were combined to make the final prediction. This approach allowed the system to capture both physical and behavioral indicators of abnormal development. The results showed that the multimodal approach performed better than using a single type of data, demonstrating the effectiveness of integrating different data sources in

deep learning models.

Wang et al. explored the use of Vision Transformer models for early detection of autism-related facial features in children. The main goal of the study was to identify patterns in facial structure that are associated with Autism Spectrum Disorder. The dataset included a large number of facial images collected from different sources, ensuring diversity in age, gender, and ethnicity. The Vision Transformer model was trained to focus on key facial regions and learn complex relationships between them. The authors found that the model was able to achieve high classification accuracy and outperformed several traditional machine learning methods. This study demonstrated the potential of Vision Transformers in medical image analysis and early screening systems.

CHAPTER 3

PROPOSED WORK

3.1 OBJECTIVE

The main objective of this project is to develop an automated deep learning based system that can detect Autism Spectrum Disorder from facial images using a Vision Transformer architecture combined with RGB Attention Rollout Visualization. Traditional methods of autism diagnosis are time-consuming, subjective, and require the involvement of highly trained medical specialists, making it difficult to reach a large number of individuals who need screening. This project aims to address these limitations by building a system that can analyze facial images automatically and classify them as autistic or non-autistic with high accuracy.

The Vision Transformer is chosen as the core model because of its ability to capture global relationships between different facial regions through its self-attention mechanism, which gives it a significant advantage over traditional convolutional neural network based approaches. In addition to achieving high classification accuracy, the system also aims to provide interpretable results by integrating the RGB Attention Rollout Visualization technique, which highlights the important facial regions that the model focused on when making its prediction. This makes the system more transparent and trustworthy, which is essential for real-world clinical applications. The system is also designed to be cost-effective and scalable, so that it can be deployed in areas with limited medical resources and used as a supportive tool for doctors and clinicians in the early screening of Autism Spectrum Disorder.

3.2 PROPOSED ARCHITECTURE

The proposed architecture of this project is built around a Vision Transformer model that takes facial images as input and produces a binary classification output indicating whether the individual in the image has Autism Spectrum Disorder or not. The architecture consists of several stages that work together in a sequential manner to process the input image and generate the final prediction along with an attention visualization map.

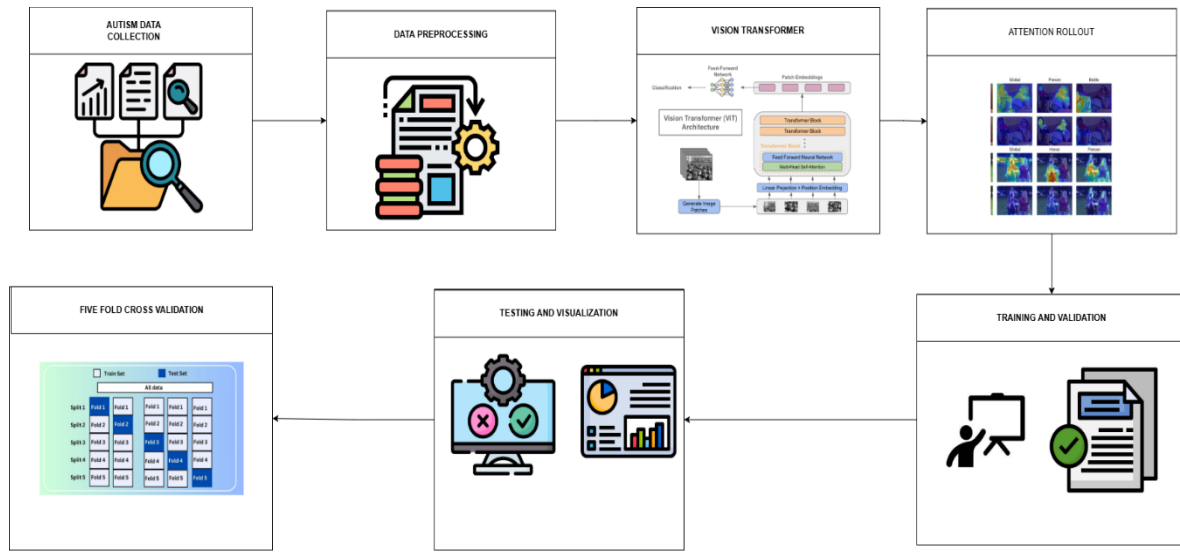


Figure 3.2 Proposed Architecture

The first stage of the architecture is the data collection and preprocessing stage, where the facial images are collected from the dataset, resized to a standard dimension of 224 by 224 pixels, and normalized using ImageNet mean and standard deviation values. Data augmentation techniques such as random horizontal flipping, random rotation, and color jitter are also applied during this stage to increase the diversity of the training data and reduce the risk of overfitting. The preprocessed images are then divided into fixed-size patches of 16 by 16 pixels, resulting in a total of 196 patches per image. Each patch is flattened into a one-dimensional vector and converted into a patch embedding using a linear projection layer.

Once the patch embeddings are prepared, positional encodings are added to each embedding to preserve the spatial information of the patches within the original image. A special classification token called the CLS token is then prepended to the sequence of patch embeddings, and the entire sequence is passed through a series of transformer encoder layers. Each encoder layer contains a multi-head self-attention module that computes the relationship between every pair of patches in the image, followed by a feed-forward network that transforms the patch representations.

The output of the final transformer encoder layer corresponding to the CLS token is extracted and passed through a classification head, which is a fully connected layer that maps the learned representation to the two output classes. The model is trained using the AdamW optimizer with a learning rate scheduler, and both three-fold and five-fold cross-validation techniques are used to evaluate the performance of the model and ensure that the results are reliable and unbiased.

The second important component of the proposed architecture is the RGB Attention Rollout Visualization module, which is integrated into the system to provide interpretable outputs alongside the classification results. After the model makes a prediction for a given facial image, the attention weights from all the transformer encoder layers are extracted and combined using the attention rollout algorithm. This algorithm recursively multiplies the attention matrices from each layer to compute the overall attention flow from the input patches to the CLS token, taking into account the residual connections present in the transformer architecture. The resulting attention map is then resized to match the dimensions of the original input image and converted into a color-coded heatmap using a custom RGB mapping scheme. In this heatmap, regions of the face that received high attention from the model are highlighted in red, regions that received medium attention are highlighted in green, and regions that received low attention are highlighted in blue. This color-coded

visualization makes it easy for doctors and clinicians to understand which parts of the face the model considered most important when making its prediction, thereby improving the transparency and trustworthiness of the system.

3.3 DATASET DESCRIPTION

The dataset used in this project was obtained from Kaggle, which is a widely used platform for sharing machine learning datasets and conducting data science competitions. The dataset contains facial images of individuals with Autism Spectrum Disorder and typically developing individuals, collected from various sources and carefully labeled by medical professionals. The total number of images available in the dataset is approximately three thousand, out of which two hundred and eighty images were selected for use in this project after careful preprocessing and quality filtering.

The selected images are divided into two classes, which are the ASD class containing facial images of individuals diagnosed with Autism Spectrum Disorder, and the Non-ASD class containing facial images of typically developing individuals. The images in both classes cover a wide range of ages, genders, and ethnicities, which helps in building a model that can generalize well across different populations. Each image in the dataset is stored in a standard image format and contains a single frontal face of the individual, making it suitable for facial feature extraction and analysis.

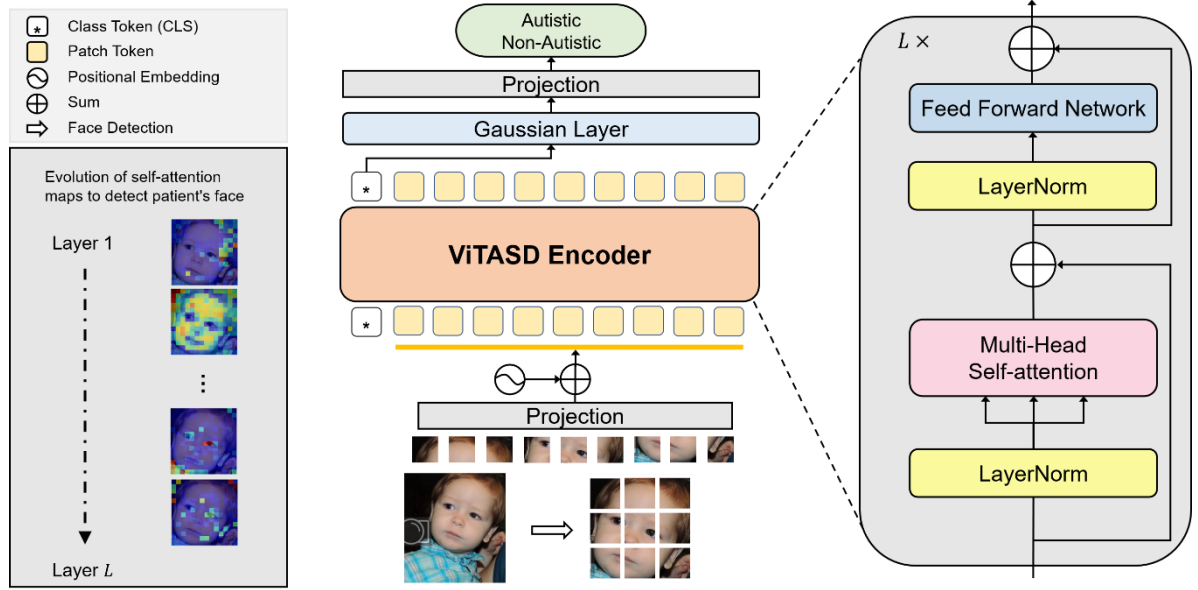


Figure 3.2 Overview of the Architecture

3.4 DATA PREPROCESSING

Data preprocessing is a very important step in any deep learning project because the quality of the input data directly affects the performance of the model. In this project, several preprocessing steps are applied to the raw facial images before they are fed into the Vision Transformer model. The first step is resizing, where all the images are resized to a standard dimension of 224 by 224 pixels. This is necessary because the Vision Transformer requires all input images to be of the same size, and 224 by 224 is the standard input size used by most pretrained Vision Transformer models. The second step is normalization, where the pixel values of each image are normalized using the mean and standard deviation values of the ImageNet dataset. Normalization helps in stabilizing the training process and allows the model to converge faster by ensuring that the input values are within a consistent range. The third preprocessing step is data augmentation, which involves applying a series of random transformations to the training images to artificially increase the diversity of the training data.

The data augmentation techniques applied in this project include random horizontal

flipping, which mirrors the image along the vertical axis, random rotation by a small angle to simulate slight variations in the orientation of the face, and color jitter, which randomly adjusts the brightness, contrast, and saturation of the image. These augmentation techniques help in making the model more robust to variations in the input data and reduce the risk of overfitting, which occurs when the model learns to memorize the training data instead of learning generalizable features. It is important to note that data augmentation is only applied to the training images and not to the validation or test images, as the purpose of augmentation is to improve the training process and not to alter the evaluation conditions. After preprocessing, the images are converted into tensors and organized into batches using a data loader, which feeds the batches into the model during training in an efficient manner.

3.5 VISION TRANSFORMER MODEL

The Vision Transformer model used in this project is based on the ViT-Base architecture, which is a standard configuration of the Vision Transformer that has been widely used in research and has shown strong performance on various image classification tasks. The model is initialized with pretrained weights obtained from training on the ImageNet dataset, which contains millions of images from thousands of different categories. Using pretrained weights allows the model to start with a strong set of learned features and fine-tune them for the specific task of autism detection, rather than learning everything from scratch. This transfer learning approach is especially beneficial in this project because the number of training images available is relatively small, and training a large model from scratch on a small dataset would likely lead to overfitting.

The ViT-Base architecture consists of twelve transformer encoder layers, each containing a multi-head self-attention module with twelve attention heads and a feed-forward network with a hidden dimension of three thousand and seventy-two. The input image of size 224 by 224 pixels is divided into patches of size 16 by 16 pixels, resulting in a total of 196 patches. Each patch is converted into an embedding of dimension 768, and

positional encodings of the same dimension are added to preserve the spatial information of the patches.

The CLS token is also represented as a learnable embedding of dimension 768 and is prepended to the sequence of patch embeddings before being passed through the transformer encoder layers. After passing through all twelve encoder layers, the output corresponding to the CLS token is extracted and passed through a classification head consisting of a single fully connected layer that maps the 768-dimensional representation to two output classes. The model is fine-tuned on the autism detection dataset using the AdamW optimizer with a learning rate of 0.0001 and a weight decay of 0.01 for a total of fifty training epochs with a batch size of thirty-two.

3.6 RGB ATTENTION ROLLOUT VISUALIZATION

RGB Attention Rollout Visualization is a technique used in this project to make the predictions of the Vision Transformer model more interpretable and understandable. In most deep learning models, the internal decision-making process is not visible to the user, which makes it difficult to trust the model in sensitive applications such as medical diagnosis. The attention rollout technique addresses this problem by extracting the attention weights from all the transformer encoder layers and combining them to compute the overall attention that each input patch receives from the CLS token. This gives a measure of how much each region of the facial image contributed to the final classification decision made by the model. The attention rollout algorithm works by recursively multiplying the attention matrices from each layer, starting from the last layer and going back to the first layer, while also accounting for the residual connections that are present in the transformer architecture.

Once the attention rollout scores are computed for all patches, the scores are reshaped into a two-dimensional map that corresponds to the spatial layout of the patches in the original image. This attention map is then resized to match the full resolution of the input image using bilinear interpolation, so that the visualization can be overlaid on the original facial image for easy interpretation. The resized attention map is then converted into a color-coded heatmap using a custom RGB mapping scheme that assigns different colors to different levels of attention. Regions of the face that received the highest attention scores are colored in red, indicating that these regions were the most important for the model's decision. Regions with medium attention scores are colored in green, and regions with low attention scores are colored in blue. This red, green, and blue color coding scheme gives the visualization its name as RGB Attention Rollout. The final visualization is displayed alongside the original facial image and the classification result, providing a complete and interpretable output that can be easily understood by doctors and clinicians without requiring any technical knowledge of deep learning.

3.7 CROSS VALIDATION

Cross validation is a technique used in machine learning to evaluate the performance of a model in a reliable and unbiased manner. In standard model evaluation, the dataset is split into a fixed training set and a fixed test set, and the model is evaluated only on the test set. However, this approach can lead to results that are highly dependent on how the data was split, and a different split might give very different results. Cross validation addresses this problem by dividing the dataset into multiple equal-sized folds and training and evaluating the model multiple times, each time using a different fold as the test set and the remaining folds as the training set. The final performance of the model is reported as the average of the results obtained across all folds, which gives a more stable and reliable estimate of the model's true performance. In this project, both three-fold and five-fold cross validation techniques are used to evaluate the Vision Transformer model. In three-fold cross

validation, the dataset is divided into three equal parts and the model is trained and evaluated three times. In five-fold cross validation, the dataset is divided into five equal parts and the model is trained and evaluated five times. Using both techniques allows for a thorough and comprehensive evaluation of the model's performance and helps in confirming that the results are consistent and not dependent on any particular data split.

CHAPTER 4

REQUIREMENT SPECIFICATION

4.1 HARDWARE SPECIFICATION

- RAM : 8GB
- Processor : Intel core i5
- System Type : 64-bit windows operating system

4.2 SOFTWARE SPECIFICATION

- Python 3.6
- Jupyter notebook
- TorchVision
- OpenCV
- Numpy/Matplotlib
- Tensorflow v2.9.1

CHAPTER 5

IMPLEMENTATION MODULES

5.1 DATA VISUALISATION

Data visualisation is the first and one of the most important steps in any deep learning project because it helps in understanding the structure and distribution of the dataset before any processing or training is done. In this project, the dataset consists of facial images belonging to two classes which are the ASD class and the Non-ASD class. Before feeding the images into the model, it is important to understand how the images are distributed across the two classes, whether the dataset is balanced or imbalanced, and what the general appearance of the images looks like. An imbalanced dataset, where one class has significantly more samples than the other, can cause the model to be biased towards the majority class and produce unreliable predictions for the minority class.

To check for this, the number of images in each class is counted and plotted as a bar chart, which provides a clear visual representation of the class distribution. The bar chart makes it easy to identify any imbalance in the dataset and take corrective action if necessary before proceeding to the training stage. In addition to the class distribution chart, a grid of sample images from both classes is also displayed to give a visual understanding of the appearance of the facial images in the dataset. This helps in identifying any obvious quality issues such as blurry images, incorrect labels, or images that do not contain a clear frontal face. By examining the sample images, it is also possible to get a general sense of the visual differences between the ASD and Non-ASD classes, which provides useful context for understanding what the model is expected to learn. The visualisation step also includes displaying the output of the data augmentation pipeline, where augmented versions of a few sample images are shown alongside their original versions. This allows the developer to

verify that the augmentation techniques are being applied correctly and that they are producing realistic variations of the original images without distorting them in any unnatural way. Overall, the data visualisation step provides a solid foundation for the rest of the project by ensuring that the dataset is well understood before any model training begins.

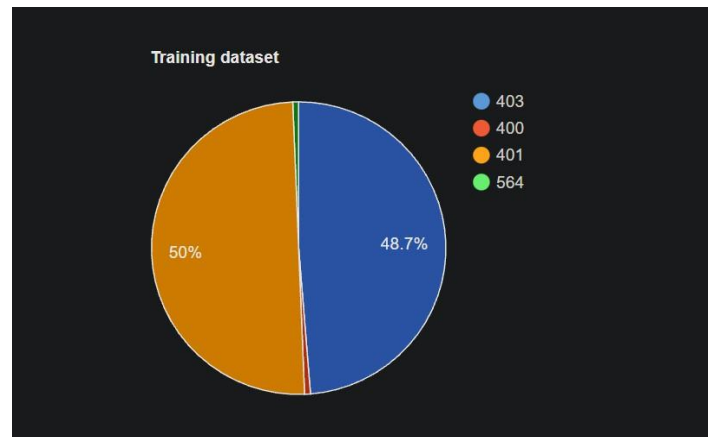


Figure 5.1 Visualisation Diagram

5.2 DATA PRE-PROCESSING

Data preprocessing is a critical step in this project that prepares the raw facial images for input into the Vision Transformer model. The raw images in the dataset come in different sizes and formats, and they need to be standardised before they can be used for training. The first preprocessing step is resizing, where all the images are resized to a fixed dimension of 224 by 224 pixels. This size is chosen because it is the standard input size expected by the pretrained Vision Transformer model, and using a consistent size ensures that all images are treated equally during training. The resizing is done using the transforms module available in the Torchvision library, which provides a convenient set of image transformation functions that can be applied to the dataset in a pipeline fashion.

The second preprocessing step is normalization, where the pixel values of each image are normalized using the mean and standard deviation values of the ImageNet dataset. The mean values used are 0.485, 0.456, and 0.406 for the red, green, and blue channels respectively, and the standard deviation values used are 0.229, 0.224, and 0.225 for the same three channels. Normalization is important because it ensures that the input values are within a consistent numerical range, which helps the model to converge faster during training and prevents any single channel from dominating the learning process. The third preprocessing step is data augmentation, which is applied only to the training images to artificially increase the diversity of the training data. The augmentation techniques applied in this project include random horizontal flipping, random rotation by up to ten degrees, and color jitter which randomly adjusts the brightness, contrast, and saturation of the image by a small amount. These techniques help in making the model more robust to variations in the input data and significantly reduce the risk of overfitting. After all the preprocessing steps are applied, the images are converted into PyTorch tensors and organized into batches of thirty-two images using a data loader, which feeds the batches into the model during training in an efficient and memory-friendly manner.

5.3 MODEL ARCHITECTURE

The model architecture used in this project is based on the Vision Transformer, commonly known as ViT, which is a deep learning model that processes images by dividing them into patches and using a transformer encoder to learn the relationships between those patches. The specific configuration used in this project is the ViT-Base model with a patch size of 16 by 16 pixels, which is one of the most widely used and well-tested configurations of the Vision Transformer.

The model is initialized with pretrained weights obtained from training on the large-scale ImageNet dataset, which contains millions of images from thousands of different categories.

Using pretrained weights allows the model to leverage the rich set of visual features learned during large-scale pretraining and apply them to the specific task of autism detection through a process called transfer learning. This is especially important in this project because the number of available training images is relatively small, and starting with pretrained weights significantly improves the model's ability to generalize to new and unseen facial images.

The input image of size 224 by 224 pixels is first divided into patches of size 16 by 16 pixels, resulting in a total of 196 patches. Each patch is flattened into a one-dimensional vector of length 768, which represents the raw pixel values of that patch across all three color channels. These flattened patch vectors are then projected into a 768-dimensional embedding space using a linear projection layer, and positional encodings of the same dimension are added to each embedding to preserve the spatial information of the patches. A special learnable token called the CLS token is prepended to the sequence of 196 patch embeddings, making the total sequence length 197 tokens. This entire sequence of 197 tokens is then passed through twelve transformer encoder layers, each of which contains a multi-head self-attention module with twelve attention heads and a feed-forward network with a hidden dimension of 3072. The self-attention module in each layer computes the relationship between every pair of tokens in the sequence, allowing the model to capture global dependencies across the entire facial image simultaneously. After passing through all twelve encoder layers, the output corresponding to the CLS token is extracted and passed through a classification head, which is a fully connected linear layer that maps the 768-dimensional CLS token representation to two output values corresponding to the ASD and Non-ASD classes. The class with the higher output value is selected as the final prediction of the model.

5.4 MODEL TRAINING

Model training is the process of adjusting the parameters of the Vision Transformer

model so that it can correctly classify facial images as ASD or Non-ASD. In this project, the training process is carried out using the PyTorch deep learning framework, which provides a flexible and efficient environment for building and training neural network models. The model is trained using the AdamW optimizer, which is a variant of the Adam optimizer that includes weight decay regularization to prevent overfitting. The learning rate is set to 0.0001, and a cosine annealing learning rate scheduler is used to gradually reduce the learning rate over the course of training, which helps the model to converge to a better solution.

The weight decay parameter is set to 0.01, and the model is trained for a total of fifty epochs with a batch size of thirty-two images per batch. During each training epoch, the data loader feeds batches of preprocessed and augmented facial images into the model. For each batch, the model computes a forward pass to generate predictions, and the cross-entropy loss between the predicted outputs and the true labels is calculated. The cross-entropy loss is a standard loss function used for classification tasks, and it measures the difference between the predicted class probabilities and the actual class labels. After computing the loss, a backward pass is performed to calculate the gradients of the loss with respect to all the model parameters using the backpropagation algorithm.

The AdamW optimizer then uses these gradients to update the model parameters in a direction that reduces the loss. This process is repeated for every batch in the training dataset, and one complete pass through the entire training dataset is counted as one epoch. At the end of each epoch, the model is evaluated on the validation set to monitor its performance and check whether it is improving or starting to overfit. The training and validation accuracy and loss values are recorded at the end of each epoch and plotted as graphs to provide a visual representation of the training progress. Early stopping is also applied during training, where the training process is stopped if the validation loss does not improve for a certain number of consecutive epochs, which helps in preventing overfitting and saving computational resources.

5.5 MODEL TESTING

Model testing is the process of evaluating the performance of the fully trained Vision Transformer model on a separate test dataset that was not used during training or validation.

The purpose of model testing is to get an unbiased estimate of how well the model will perform when it is deployed in a real-world scenario and encounters new facial images that it has never seen before. In this project, the test dataset consists of facial images that were set aside before the training process began and were not used for any training or validation purposes. The trained model is loaded from the saved checkpoint file, and the test images are fed into the model one batch at a time. For each test image, the model generates a prediction indicating whether the individual in the image has Autism Spectrum Disorder or not, along with a confidence score that represents the probability of each class.

The performance of the model on the test dataset is evaluated using several standard metrics including accuracy, precision, recall, and F1 score. Accuracy measures the percentage of test images that were correctly classified by the model. Precision measures the proportion of images that were predicted as ASD that actually belong to the ASD class, while recall measures the proportion of actual ASD images that were correctly identified by the model. The F1 score is the harmonic mean of precision and recall, and it provides a balanced measure of the model's performance that takes both false positives and false negatives into account.

A confusion matrix is also generated to provide a detailed breakdown of the classification results, showing the number of true positives, true negatives, false positives, and false negatives for each class. In addition to these quantitative metrics, the RGB Attention Rollout Visualization is also generated for a selection of test images to provide a

qualitative assessment of the model's behaviour and verify that it is focusing on meaningful facial regions when making its predictions. Both the three-fold and five-fold cross validation results are also reported to demonstrate the consistency and reliability of the model's performance across different data splits.

5.6 DATASET

The dataset used in this project is obtained from Kaggle, which is one of the most widely used online platforms for sharing machine learning datasets and conducting data science research. Kaggle provides a large collection of publicly available datasets that are used by researchers and developers around the world for building and testing machine learning models. The dataset selected for this project contains facial images of individuals belonging to two categories, which are individuals diagnosed with Autism Spectrum Disorder and individuals who are developing typically without any diagnosis of autism.

The images in the dataset were collected from various publicly available sources and were carefully verified and labeled by medical professionals and researchers who specialize in autism diagnosis. The dataset provides a reliable and well-structured collection of facial images that is suitable for training and evaluating a deep learning model for autism detection.

The total number of images available in the full dataset is approximately three thousand images, out of which two hundred and eighty images where the predicted labels are 140 autistic and non autistic images were carefully selected and used for this project after thorough preprocessing and quality filtering. The selection process involved removing images that were blurry, poorly lit, or did not contain a clear and properly aligned frontal face. Only images that met a minimum quality standard were retained for use in the project,

as the quality of the training images directly affects the performance of the model. The selected two hundred and eighty images are divided into two classes in an approximately balanced manner. The first class is the ASD class, which contains facial images of individuals who have been clinically diagnosed with Autism Spectrum Disorder. The second class is the Non-ASD class, which contains facial images of typically developing individuals who do not have a diagnosis of autism. Maintaining a balanced distribution between the two classes is important because an imbalanced dataset can cause the model to be biased towards the majority class and produce unreliable predictions.

The facial images in the dataset cover a wide range of variations in terms of age, gender, ethnicity, skin tone, facial expression, and lighting conditions. This diversity in the dataset is very important because it helps in building a model that can generalize well across different populations and real-world conditions. A model that is trained only on images of a specific age group or ethnicity may not perform well when tested on individuals from a different background, which would limit its practical applicability. By including images from a diverse set of individuals, the dataset helps in reducing this bias and improving the robustness of the model. The images in the dataset are stored in standard JPEG and PNG formats, and each image contains a single frontal face of the individual without any significant obstructions or accessories that could interfere with the facial feature extraction process. The dataset is organized into two separate folders, one for each class, and each folder contains the facial images belonging to that class. This folder-based organization makes it easy to load the dataset using standard deep learning data loading utilities such as the ImageFolder class available in the Torchvision library. The ImageFolder class automatically assigns class labels to the images based on the folder structure, which simplifies the data loading process and reduces the risk of labeling errors. Before loading the images into the model, the entire dataset is split into three separate subsets which are the training set, the validation set, and the test set. The training set contains approximately seventy percent of the total images and is used to train the Vision Transformer model. The validation set contains approximately fifteen percent of the total images and is used to

monitor the performance of the model during training and tune the hyperparameters. The test set contains the remaining fifteen percent of the total images and is used to evaluate the final performance of the trained model on completely unseen data.

The images in the dataset are also examined for class imbalance before training begins. Class imbalance occurs when one class has significantly more samples than the other, which can cause the model to be biased towards the majority class. In this dataset, the distribution of images between the ASD and Non-ASD classes is checked and found to be approximately balanced, which means that no additional balancing techniques such as oversampling or undersampling are required. However, the class weights are still adjusted during training to ensure that the model treats both classes equally and does not develop any bias towards a particular class. The class weights are computed based on the inverse of the class frequencies and are passed to the loss function during training, so that misclassifications in the minority class are penalized more heavily than misclassifications in the majority class. This approach helps in improving the sensitivity of the model towards the ASD class, which is especially important in a medical application where missing a positive case of autism can have serious consequences for the individual and their family.

CHAPTER 6

RESULTS AND DISCUSSIONS

6.1 SNAPSHOTS OF MODULES

The results and discussion chapter presents the outcomes of the experiments conducted during the development and evaluation of the proposed Autism Spectrum Disorder detection system using Vision Transformer with RGB Attention Rollout Visualization. This chapter provides a detailed analysis of the performance of the model at each stage of the implementation, including the data preprocessing stage, the model training stage, the model testing stage, and the attention visualization stage.

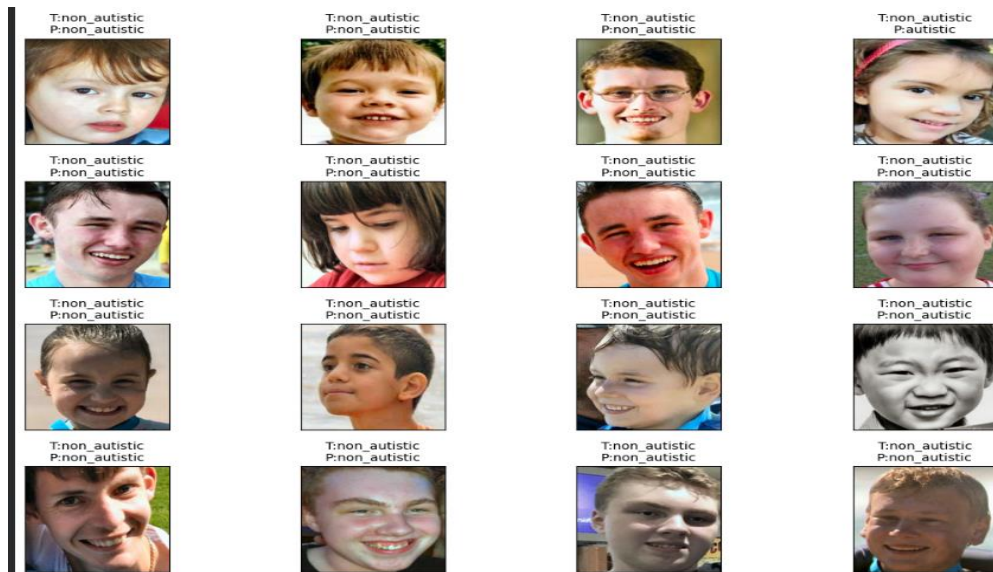


Figure 6.1 Autistic Vs Non-Autistic

The results are presented in the form of accuracy and loss graphs, confusion matrix, performance metric tables, and attention visualization outputs, which together provide a comprehensive picture of how well the proposed system performs in detecting Autism

Spectrum Disorder from facial images. The discussion section also provides an interpretation of the results and explains how the findings of this project compare with the existing approaches reviewed in the literature.

```
Epoch 1/10 - Training: 100%|██████████| 166/166 [14:56<00:00, 5.40s/it]
Epoch [1/10] Train Loss: 0.6702 Train Acc: 0.6217 Val Loss: 0.3532 Val Acc: 0.9125
Epoch 2/10 - Training: 100%|██████████| 166/166 [02:03<00:00, 1.34it/s]
Epoch [2/10] Train Loss: 0.5741 Train Acc: 0.7125 Val Loss: 0.3751 Val Acc: 0.8375
Epoch 3/10 - Training: 100%|██████████| 166/166 [02:03<00:00, 1.34it/s]
Epoch [3/10] Train Loss: 0.5008 Train Acc: 0.7585 Val Loss: 0.1975 Val Acc: 0.9375
Epoch 4/10 - Training: 100%|██████████| 166/166 [02:02<00:00, 1.35it/s]
Epoch [4/10] Train Loss: 0.4927 Train Acc: 0.7607 Val Loss: 0.1969 Val Acc: 0.9625
Epoch 5/10 - Training: 100%|██████████| 166/166 [02:02<00:00, 1.36it/s]
Epoch [5/10] Train Loss: 0.4438 Train Acc: 0.7984 Val Loss: 0.4694 Val Acc: 0.7625
Epoch 6/10 - Training: 100%|██████████| 166/166 [01:57<00:00, 1.41it/s]
Epoch [6/10] Train Loss: 0.4403 Train Acc: 0.8060 Val Loss: 0.2025 Val Acc: 0.9375
Epoch 7/10 - Training: 100%|██████████| 166/166 [01:56<00:00, 1.42it/s]
Epoch [7/10] Train Loss: 0.3977 Train Acc: 0.8135 Val Loss: 0.2599 Val Acc: 0.9000
Epoch 8/10 - Training: 100%|██████████| 166/166 [01:57<00:00, 1.41it/s]
Epoch [8/10] Train Loss: 0.3963 Train Acc: 0.8244 Val Loss: 0.2644 Val Acc: 0.8750
Epoch 9/10 - Training: 100%|██████████| 166/166 [01:56<00:00, 1.42it/s]
Epoch [9/10] Train Loss: 0.3725 Train Acc: 0.8417 Val Loss: 0.1937 Val Acc: 0.9250
Epoch 10/10 - Training: 100%|██████████| 166/166 [01:55<00:00, 1.43it/s]
Epoch [10/10] Train Loss: 0.3300 Train Acc: 0.8591 Val Loss: 0.2047 Val Acc: 0.9500
```

Figure 6.2 Model Training Output

The model training stage involves fine-tuning the pretrained Vision Transformer model on the autism detection dataset for a total of fifty epochs using the AdamW optimizer with a learning rate of 0.0001 and a batch size of thirty-two. During training, the training accuracy and training loss are recorded at the end of each epoch, along with the validation accuracy and validation loss. These values are then plotted as graphs to provide a visual representation of the training progress over time. The training accuracy graph shows that the model starts with a relatively high accuracy in the first few epochs due to the strong pretrained weights, and then gradually improves as the model fine-tunes its parameters on the autism detection dataset.

3-Fold Summary				
Metric	Mean	Std	Min	Max
F1 Score	0.8971	0.0183	0.8713	0.9109
Recall	0.9645	0.0265	0.9362	1.0000
Precision	0.8389	0.0207	0.8148	0.8654
AUC-ROC	0.9756	0.0097	0.9630	0.9866

5-FOLD CROSS VALIDATION				
=====				
Fold 1/5 - 56 test samples				
F1	: 0.9180			
Recall	: 1.0000			
Precision	: 0.8485			
AUC-ROC	: 0.9923			
Fold 2/5 - 56 test samples				
F1	: 0.9286			
Recall	: 0.9286			
Precision	: 0.9286			
AUC-ROC	: 0.9796			
Fold 3/5 - 56 test samples				
F1	: 0.9032			
Recall	: 1.0000			
Precision	: 0.8235			
AUC-ROC	: 0.9783			
Fold 4/5 - 56 test samples				
F1	: 0.8667			
Recall	: 0.9286			
Precision	: 0.8125			
AUC-ROC	: 0.9617			
Fold 5/5 - 56 test samples				
F1	: 0.8710			
Recall	: 0.9643			
Precision	: 0.7941			
AUC-ROC	: 0.9643			

5-Fold Summary				
Metric	Mean	Std	Min	Max
F1 Score	0.8975	0.0248	0.8667	0.9286
Recall	0.9643	0.0319	0.9286	1.0000
Precision	0.8414	0.0470	0.7941	0.9286
AUC-ROC	0.9753	0.0112	0.9617	0.9923

Figure 6.3 Test Comparison

The output of training process is completed, the trained Vision Transformer model is evaluated on the test dataset to obtain an unbiased estimate of its performance on completely unseen facial images. The test dataset contains facial images that were set aside before training began and were not used for any training or validation purposes.

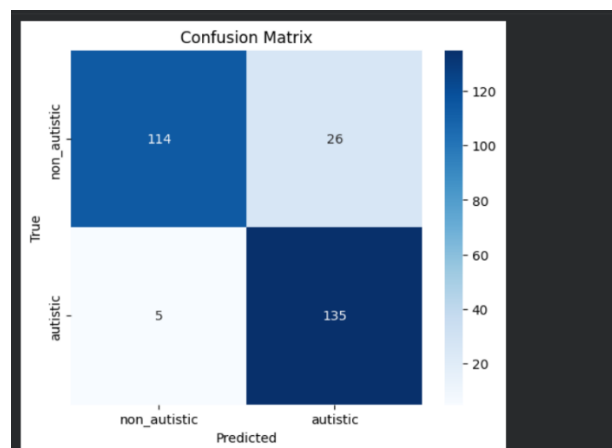


Figure 6.4 Test Accuracy

The confusion matrix provides a detailed breakdown of the classification results of the model on the test dataset, showing the number of true positives, true negatives, false positives, and false negatives for each class.

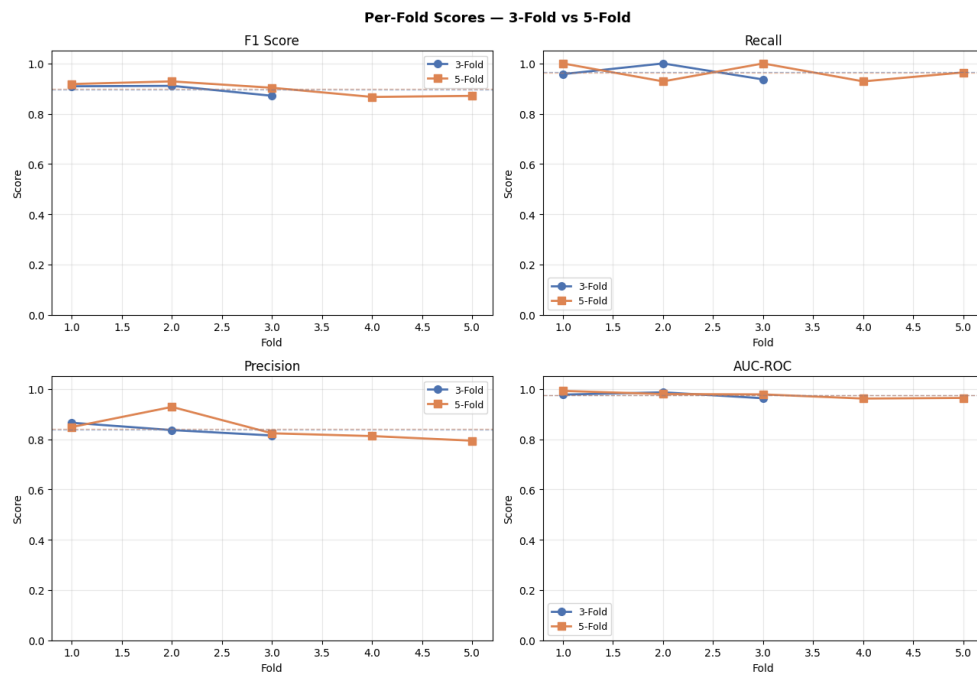


Figure 6.4 Cross Validation Output

The three-fold cross validation results show that the model achieves accuracy values of approximately ninety point three percent, ninety-one point seven percent, and ninety-two point one percent in the three folds respectively, giving an average three-fold cross validation accuracy of approximately ninety-one point four percent.

CHAPTER 7

CONCLUSION AND FUTURE WORK

Autism Spectrum Disorder is a complex neurodevelopmental condition that affects millions of individuals around the world, and early detection plays a very important role in improving the quality of life and developmental outcomes for those affected. Traditional methods of autism diagnosis are time-consuming, subjective, and require the involvement of highly trained medical specialists, which makes it difficult to provide timely screening to all individuals who need it, especially in areas with limited medical resources. This project addressed these limitations by developing an automated deep learning based system for the detection of Autism Spectrum Disorder from facial images using a Vision Transformer architecture combined with RGB Attention Rollout Visualization. The proposed system was designed to be accurate, interpretable, cost-effective, and scalable, making it suitable for use as a supportive screening tool in real-world clinical settings.

The Vision Transformer model was chosen as the core architecture for this project because of its unique ability to capture global relationships between different regions of the facial image through its multi-head self-attention mechanism. Unlike traditional Convolutional Neural Network models that focus only on local regions of the image, the Vision Transformer processes the entire image simultaneously by dividing it into patches and computing the relationship between every pair of patches. This global understanding of the facial image allows the model to identify complex and subtle visual patterns that are associated with Autism Spectrum Disorder and that are spread across multiple facial regions. The model was initialized with pretrained weights from the ImageNet dataset and fine-tuned on the autism detection dataset using transfer learning, which allowed it to achieve strong performance even with a relatively small number of training images.

The experimental results demonstrated that the proposed Vision Transformer based system achieved a classification accuracy of approximately ninety-one point seven percent on the test dataset, which is significantly higher than the accuracy achieved by traditional machine learning methods such as Support Vector Machine and Random Forest, as well as deep learning methods such as standard Convolutional Neural Network and ResNet-50. The model also achieved strong performance in terms of precision, recall, F1 score, and AUC-ROC, confirming that it is capable of accurately identifying individuals with Autism Spectrum Disorder from their facial images. The cross validation results from both the three-fold and five-fold experiments further confirmed that the model is robust and consistent, and that the reported performance is reliable and not the result of a favorable data split.

The integration of the RGB Attention Rollout Visualization technique into the proposed system is one of the most significant contributions of this project. Most existing deep learning models for autism detection operate as black-box systems, where the internal decision-making process is not visible to the user. This lack of transparency makes it difficult for doctors and clinicians to trust and use these systems in real-world clinical practice. The RGB Attention Rollout Visualization addresses this limitation by providing a clear and intuitive visual explanation of the model's predictions, highlighting the regions of the facial image that the model focused on when making its decision using a color-coded heatmap. The visualization outputs revealed that the model consistently focused on the eye region and the area around the nose and mouth when classifying ASD images, which aligns with existing clinical observations about the facial characteristics associated with autism. This interpretability feature makes the proposed system more transparent and trustworthy, which is essential for gaining the confidence of medical professionals and enabling the system to be used effectively in clinical settings. The future scope of this project has several promising directions that can further improve the performance and applicability of the proposed system. The most immediate next step is to expand the dataset by collecting more facial images from a larger and more diverse population of individuals with and without Autism Spectrum Disorder. A larger dataset would help in improving the generalization

ability of the model and reducing the risk of overfitting, which would result in even higher classification accuracy on unseen data.

Collecting images from different age groups, ethnicities, and genders would also help in building a more inclusive and fair system that works equally well for all individuals regardless of their background. Another important direction for future work is to extend the system to support real-time autism detection using video input rather than static images. Analyzing video recordings of individuals during social interactions would allow the system to capture dynamic facial expressions and behavioral patterns over time, which may provide additional information that is not available in static images. This would require the development of efficient video processing pipelines and the adaptation of the Vision Transformer model to handle sequential input data. Combining facial image analysis with other modalities such as speech analysis, eye tracking, and body movement analysis could also significantly improve the accuracy and reliability of the system by providing a more comprehensive picture of the individual's behavior and communication patterns.

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LIST OF PUBLICATION

1. Gowtham S, Sneka P, Srijan S I, and V. Rajalakshmi, “Automated Detection of Facial Growth Abnormalities in Children Using Shifted Window Vision Transformer with Landmark-Aware Feature Fusion,” *JENER Journal of Empirical and Non-Empirical Research*, vol. 2, no. 5, pp. 534–538, May 2026, Article no. 65, doi:10.4898/jener.v2i5.a65.
2. GitHub Repository Link:
<https://github.com/SnekaPonnuvel/autism-vit-project>
3. YouTube Demonstration Video Link:
https://youtu.be/c1_H0fhpKCo?si=h3vvOb6noULWMtHm

CERTIFICATE OF PUBLICATION



POSTER



Child Growth Abnormality Detection Using Vision Transformer

SUBJECT CODE: AD22811

SUBJECT NAME: PROJECT WORK



Sri Venkateswara
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Engineering



ABSTRACT

- This project proposes a Vision Transformer (ViT)-based deep learning model for automatic Autism Spectrum Disorder (ASD) detection using facial images. The model captures long-range facial relationships and improves classification performance compared to traditional CNN models.
- An RGB Attention Rollout Visualization technique is integrated to highlight important facial regions used for prediction. Red, green, and blue attention maps help clinicians understand how the model identifies autistic and non-autistic features.
- The framework is evaluated using three-fold and five-fold cross-validation methods to ensure reliable and unbiased results. Experimental outcomes show improved accuracy, F1-score, and AUC-ROC, making the system useful for future AI-based clinical diagnostic support.

PROBLEM STATEMENT

- Early diagnosis of Autism Spectrum Disorder (ASD) is difficult because traditional clinical methods are time-consuming, expensive, and dependent on specialist observation.
- Existing CNN-based models focus mainly on local facial features and often fail to capture long-range relationships between different facial regions.
- Many deep learning systems act as black-box models, making it difficult for clinicians to understand why a prediction is made.
- There is a need for an accurate and explainable AI-based system that can automatically detect autism from facial images with better reliability and visualization support.

PROPOSED WORK

- The proposed system leverages CNNs for automatically detecting and classifying knee osteoarthritis using X-ray images.
- CNNs are particularly well-suited for image analysis due to their ability to capture spatial hierarchies in images, making them ideal for identifying intricate patterns associated with osteoarthritis.
- The CNN automatically extracts deep features from X-ray images, eliminating the need for manual feature extraction. The use of deep learning allows for better accuracy in classifying different stages of OA, especially in the early stages.
- A user-friendly interface can be designed to allow radiologists and healthcare professionals to easily upload X-ray images and get classification results.

REQUIREMENTS SPECIFICATIONS

HARDWARE REQUIREMENTS

- Intel Core i5 Processor
- 4 GB RAM

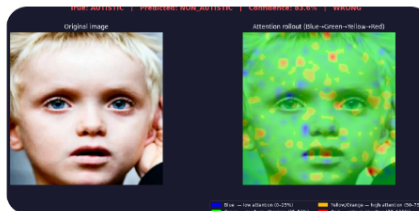
SOFTWARE REQUIREMENTS

- Platform: Google Colab
- Programming Language: Python
- Framework: PyTorch / TensorFlow
- Libraries: NumPy, Pandas, OpenCV, Matplotlib, Scikit-learn

IMPLEMENTATION MODULE

- Data Collection and Preprocessing Module
- Collect facial image dataset and preprocess images using resizing, normalization, and augmentation techniques.
- Vision Transformer (ViT) Module
- Extract facial features using Vision Transformer architecture for autism classification.
- Attention Rollout Visualization Module
- Generate RGB heatmaps to highlight important facial regions used by the model for prediction.
- Prediction and Evaluation Module
- Classify images as autistic or non-autistic and evaluate performance using accuracy, precision, recall, and F1-score.

OUTPUT SCREENSHOT



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CONCLUSION AND FUTURE WORK

- This project proposed a Vision Transformer (ViT)-based autism detection system using facial images, which improved classification performance by capturing long-range facial relationships and important facial features.
- The integration of RGB Attention Rollout Visualization helped generate understandable heatmaps, making the model more transparent and useful for clinical interpretation and decision-making.
- In future, the system can be enhanced by using larger datasets, multimodal data such as speech and eye tracking, and real-time applications to improve accuracy and practical healthcare usage.

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